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Architecture Models – Complete Reference

Parent Model: architecture-model-standard

- **Schema Version:** 2.0
- **Validation Score:** 100/100
- **Components:** 25
- **Capabilities:** 10
- **Interfaces:** 7
- **Behaviors:** 5
- **Constraints:** 2
- **Layers:** 1
- **Actors:** 2
- **Relationships:** 93
- **Signatures:** 85
- **Constants:** 26
- **Test Contracts:** 266

Actors

- **ACT-DEV** – Developer: Software engineer using CLI to manage architecture models
- **ACT-LLM** – LLM Agent: AI agent consuming architecture models via opencode-arch MCP tools

Capabilities

ID	Name	F-Block	Description
CAP-F1	Model Parsing & Validation	F3	Parse YAML models and validate structural correctness (score 0-100)

ID	Name	F-Block	Description
CAP-F2	Reality Manifest Generation	F5	AST scanning to produce ground-truth code inventories with metrics, body hints, and test contracts
CAP-F3	Model Slicing & Diffing	F3	Subset extraction by F-block/layer, model version comparison, and enriched model composition
CAP-F4	CLI Operations	F1	Command-line interface for init, validate, slice, diff, stats, impact, manifest, coverage, enrich
CAP-F5	Configuration Management	F2	Auto-discovery config loading and project configuration schema
CAP-F6	Schema Specification	F8	JSON Schema definitions for architecture model validation
CAP-F7	Model Extraction	F4	Extract architecture model from source code
CAP-F8	Domain Profiles	F7	Cross-domain architecture modeling via domain profiles (software, controls, mechanical, electrical)
CAP-F9	Shared Utilities	F9	File discovery, exclusion patterns, and shared helper functions
CAP-F10	Auto-Enrichment	F6	Populate signatures, constants, and test_contracts from AST scanning

Interfaces

ID	Name	Type	Description
IF-CLI	CLI Interface	InterfaceType.INTERNAL	Entry point: architecture-model {init,validate,slice,diff,stats,impact,manifest,coverage,enrich}
IF-PARSE-API	Parser API	InterfaceType.INTERNAL	load_model(path) -> ArchitectureModel, save_model(model, path), dump_model(model) -> str
IF-VALIDATE-API	Validator API	InterfaceType.INTERNAL	validate_model(model) -> ValidationResult (score, issues, is_valid)
IF-MANIFEST-API	Manifest API	InterfaceType.INTERNAL	generate_manifest(project_root) -> Manifest
IF-SLICER-API	Slicer API	InterfaceType.INTERNAL	slice_by_fblock(model, id), slice_by_layer(model, id), slice_for_artifact(model, artifact)
IF-PROFILE-API	Profile API	InterfaceType.INTERNAL	load_profile(name) -> DomainProfile
IF-ENRICH-API	Enrichment API	InterfaceType.INTERNAL	enrich_model(model, project_root) -> ArchitectureModel

Behaviors

- **BEH-INIT** – Project Initialization: Scan directory structure, discover source root, auto-generate .architecture-model.yaml
- **BEH-VALIDATE** – Model Validation: Check architectural claims against schema and structural invariants, produce score 0-100

- **BEH-MANIFEST** – Manifest Generation: Walk source tree, parse ASTs, extract functions/classes/imports, compute metrics, discover blocks
- **BEH-ENRICH** – Auto-Enrichment: Enrich architecture model with function signatures, constants, and test contracts from AST
- **BEH-EXTRACT** – Model Extraction from Code: Extract architecture model entities and relationships from source code analysis

Constraints

- **CON-SCHEMA** – Schema Compliance (ConstraintType.TECHNOLOGY): All models must conform to JSON Schema v1.5 with 7 entity types and 8 relationship types
- **CON-NO-ORPHANS** – No Orphaned Entities (ConstraintType.TECHNOLOGY): Every entity must participate in at least one relationship

Components

ID	Name	F-Block	Sigs	Const	Contracts	Files	Regen
COMP- CLI	cli	F1	1	0	0	2	NO
COMP- CONFIG	config	F2	15	1	1	3	YES
COMP- CORE	core	F3	0	0	0	0	NO
COMP- CORE-DECOMPOSER	core.decomposer	F3	5	1	15	1	YES
COMP- CORE-DIFFER	core.differ	F3	8	0	0	1	NO
COMP- CORE-MERGER	core.merger	F3	8	5	2	1	YES
COMP- CORE-PARSER	core.parser	F3	4	1	11	1	YES
COMP- CORE-SLICER	core.slicer	F3	4	1	12	1	YES
COMP- CORE-TYPES	core.types	F3	6	0	30	1	YES
COMP- CORE-VALIDATOR	core.validator	F3	7	2	11	1	YES
COMP- DECOMPOSE	decompose	F6	2	0	3	1	YES
COMP- ENRICH	enrich	F6	1	0	2	1	YES
COMP- EXTRACT	extract	F4	1	0	0	1	NO
COMP- MANIFEST	manifest	F5	0	0	0	0	NO

ID	Name	F-Block	Sigs	Const	Contracts	Files	Regen
COMP- MANIFEST- BLOCKS	manifest.blocks	F5	1	1	21	1	YES
COMP- MANIFEST- BODY- HINTS	manifest.body_hints	F5	3	4	29	1	YES
COMP- MANIFEST- GENERATOR	manifest.generator	F5	4	1	21	3	YES
COMP- MANIFEST- INTERFACES	manifest.interfaces	F5	1	1	20	1	YES
COMP- MANIFEST- METRICS	manifest.metrics	F5	1	1	19	1	YES
COMP- MANIFEST- SCANNER	manifest.scanner	F5	1	2	21	1	YES
COMP- MANIFEST- TEST- ANALYZER	manifest.test_analyzer	F5	2	3	24	1	YES
COMP- MANIFEST- TYPES	manifest.types	F5	3	1	17	1	YES
COMP- PROFILES	profiles	F7	3	1	0	3	NO
COMP- SPEC	spec	F8	0	0	0	1	NO
COMP- UTILS	utils	F9	4	0	7	2	YES

Component Details

COMP-CLI: cli

- **F-Block:** F1
- **Description:** CLI entry point and command dispatch (9 commands: init, validate, slice, diff, stats, impact, manifest, coverage, enrich)
- **Files:** src/architecture_model/cli/main.py, src/architecture_model/cli/init.py

Signatures:

- main(argv: list[str] | None) --> int

COMP-CONFIG: config

- **F-Block:** F2
- **Description:** Configuration loader (auto-discovery) and schema definition
- **Files:** src/architecture_model/config/loader.py, src/architecture_model/config/schema.py, src/architecture_model/config/

Constants:

- `CONFIG_FILENAME = .architecture-model.yaml`

Signatures:

- `load_config(root: Path) --> ProjectConfig`
- `discover_config(root: Path) --> tuple[ProjectConfig, DiscoveryReport]`
- `get_config(root: Path) --> ProjectConfig`
- `write_config(config: ProjectConfig, root: Path | None) --> Path`
- `resolve(project_name: str, root: Path) --> 'ResolvedOutputConfig'`
- `layer_dir_map() --> dict[str, list[str]] @property`
- `fblock_dir_map() --> dict[str, str] @property`
- `fblock_dict() --> dict[str, dict[str, Any]] @property`
- `metrics_paths() --> dict[str, Path] @property`
- `resolved_output() --> ResolvedOutputConfig`
- `from_dict(data: dict[str, Any], root: Path) --> 'ProjectConfig' @classmethod`
- `to_dict() --> dict[str, Any]`
- `add_candidate(category: str, path: str, accepted: bool, reason: str) --> None`
- `claim_rate() --> float @property`
- `summary() --> str`

Test Contracts (1):

Test File	Method	Assertion	Type
test_config_discovery_typed.py	test_discover_config_returns_repo	<code>config_repo == tmp_path.name</code>	value_equality

COMP-CORE: core

- **F-Block:** F3
- **Description:** Parser, validator, slicer, differ, merger, decomposer, type system

COMP-CORE-DECOMPOSER: core.decomposer

- **F-Block:** F3
- **Description:** Model decomposition into subsystems
- **Files:** `src/architecture_model/core/decomposer.py`

Constants:

- `SYSTEM_THRESHOLD = 10.0`

Signatures:

- `compute_complexity(comp: Component, model: ArchitectureModel) --> float`
- `identify_systems(model: ArchitectureModel, manifest: dict) --> list[SystemCandidate]`
- `auto_assign_f_blocks(model: ArchitectureModel, max_cluster_size: int) --> ArchitectureModel`
- `decompose_model(model: ArchitectureModel, manifest: dict, output_dir: str) --> DecompositionResult`
- `test_affinity_decompose(repo_path: Path) --> list[Subsystem]`

Test Contracts (15):

Test File	Method	Assertion	Type
test_decomposer.py	test_empty_component_compute_complexity	<code>compute_complexity(comp, model) == 0.0</code>	value_equality

Test File	Method	Assertion	Type
test_decomposer.py	test_symbols_contributescore	== pytest.approx(4.9)	value_equality
test_decomposer.py	test_functions_contributescore	== pytest.approx(2.0)	value_equality
test_decomposer.py	test_deps_contribute	score == pytest.approx(3.0)	value_equality
test_decomposer.py	test_complex_component_score	== pytest.approx(36.0)	value_equality
test_decomposer.py	test_non_depends_on_relation	== 0	value_equality
test_decomposer.py	test_identifies_complex_block_systems	== 1	value_equality
test_decomposer.py	test_simple_model_no_systems	== 0	value_equality
test_decomposer.py	test_components_without_systems_grouped	== 0	value_equality
test_decomposer.py	test_multiple_systems_identified	== 2	value_equality
test_decomposer.py	test_uses_fblock_id_as_key(systems) not in manifest		value_equality
test_decomposer.py	test_decompose_creates_result_instance		type_check
		DecompositionResult	
test_decomposer.py	test_inter_system_rels_defined(simple)	== 1	value_equality
test_decomposer.py	test_sub_model_meta_has_system_name	== 'Core Engine'	value_equality
test_decomposer.py	test_no_systems_returns_unreduced_level.entities.systems	== 0	value_equality

COMP-CORE-DIFFER: `core.differ`

- **F-Block:** F3
- **Description:** Model version comparison and diff generation
- **Files:** `src/architecture_model/core/differ.py`

Signatures:

- `has_changes()` --> `bool @property`
- `added_count()` --> `int @property`
- `removed_count()` --> `int @property`
- `modified_count()` --> `int @property`
- `summary()` --> `str`
- `format_report()` --> `str`
- `affected_artifacts()` --> `set[str]`
- `diff_models(old_model: ArchitectureModel, new_model: ArchitectureModel) --> ModelDiff`

COMP-CORE-MERGER: `core.merger`

- **F-Block:** F3
- **Description:** Model merging and composition
- **Files:** `src/architecture_model/core/merger.py`

Constants:

- `_MAX_MEMBERS_PER_SYMBOL = 8`
- `_MAX_FUNCTIONS_PER_COMPONENT = 12`
- `_MAX_SYMBOLS_PER_COMPONENT = 6`
- `_YAML_CHAR_BUDGET = 12000`
- `_EXCLUDED_FILES = frozenset({'setup.py', 'conftest.py'})`

Signatures:

- `merge_manifest(model: ArchitectureModel, manifest_path: str | Path, project_root: str | Path | None) --> ArchitectureModel`
- `model()` --> `ArchitectureModel @property`

- `entities()` @property
- `relationships()` @property
- `meta()` @property
- `enrich_from_manifest(model: ArchitectureModel, manifest: dict) --> EnrichmentResult`
- `compact_for_generation(model: ArchitectureModel) --> ArchitectureModel`
- `compose_enriched_model(project_root: Path) --> ArchitectureModel`

Test Contracts (2):

Test File	Method	Assertion	Type
test_merger.py	test_manifest_hash_updated	<code>load(model_copy.meta.manifest_hash) == 16</code>	value_equality
test_merger.py	test_missing_manifest_reloaded	<code>model_copy.manifest_hash == original_hash</code>	value_equality

COMP-CORE-PARSER: `core.parser`

- **F-Block:** F3
- **Description:** YAML model parser (`load_model`, `save_model`, `dump_model`)
- **Files:** `src/architecture_model/core/parser.py`

Constants:

- `SCHEMA_PATH = Path(__file__).parent.parent / 'spec' / 'schema.json'`

Signatures:

- `load_model(path: str | Path) --> ArchitectureModel`
- `validate_model_data(data: dict[str, Any]) --> list[str]`
- `dump_model(model: ArchitectureModel) --> dict[str, Any]`
- `save_model(model: ArchitectureModel, path: str | Path) --> None`

Test Contracts (11):

Test File	Method	Assertion	Type
test_parser.py	test_load_returns_architecture_model_instance	<code>model is instance ArchitectureModel</code>	type_check
test_parser.py	test_meta_has_schema_version	<code>model.meta.schema_version == '0.1.0'</code>	value_equality
test_parser.py	test_empty_file_raises	raises <code>ValueError</code>	raises
test_parser.py	test_missing_file_raises	raises <code>FileNotFoundError</code>	raises
test_parser.py	test_malformed_yaml_raises	raises <code>Exception</code>	raises
test_parser.py	test_round_trip_preserves_entity_count	<code>reloaded.entity_count == model.entity_count</code>	value_equality
test_parser.py	test_round_trip_preserves_relationship_count	<code>reloaded.relationship_count == model.relationship_count</code>	value_equality
test_parser.py	test_round_trip_preserves_project	<code>reloaded.meta.project == model.meta.project</code>	value_equality
test_parser.py	test_round_trip_preserves_ids	<code>original_ids == reloaded_ids</code>	value_equality
test_parser.py	test_real_model_passes_schema_errors	<code>len(errors) == 0</code>	value_equality
test_parser.py	test_minimal_valid_structured	<code>tree_errors is instance list</code>	type_check

COMP-CORE-SLICER: `core.slicer`

- **F-Block:** F3
- **Description:** Model subsetting by F-block, layer, or artifact
- **Files:** `src/architecture_model/core/slicer.py`

Constants:

- `ACTIVE = ACTIVE` (Status enum member)

Signatures:

- `slice_by_fblock(model: ArchitectureModel, f_block: str, include_relationships: bool) --> ArchitectureModel`
- `slice_by_layer(model: ArchitectureModel, layer_id: str) --> ArchitectureModel`
- `slice_by_status(model: ArchitectureModel, status: Status) --> ArchitectureModel`
- `slice_for_artifact(model: ArchitectureModel, artifact_name: str) --> ArchitectureModel`

Test Contracts (12):

Test File	Method	Assertion	Type
<code>test_slicer.py</code>	<code>test_returns_only_fblock_capabilities</code>	<code>sliced.capabilities == 'F1'</code>	value_equality
<code>test_slicer.py</code>	<code>test_returns_only_fblock_components</code>	<code>sliced.components == 'F2'</code>	value_equality
<code>test_slicer.py</code>	<code>test_nonexistent_fblock_returns_empty_capabilities</code>	<code>len(sliced.capabilities) == 0</code>	value_equality
<code>test_slicer.py</code>	<code>test_no_relationship_model</code>	<code>sliced.relationship_count == 0</code>	value_equality
<code>test_slicer.py</code>	<code>test_registered_artifact_sliced_is_instance</code>	<code>sliced.isinstance(ArchitectureModel)</code>	type_check
<code>test_slicer.py</code>	<code>test_registered_artifact_preserves_project</code>	<code>sliced.project == model.meta.project</code>	value_equality
<code>test_slicer.py</code>	<code>test_functional_has_no_components</code>	<code>len(sliced.entities.components) == 0</code>	value_equality
<code>test_slicer.py</code>	<code>test_unregistered_artifact_sliced_is_full_copy</code>	<code>sliced.entity_count == model.entity_count</code>	value_equality
<code>test_slicer.py</code>	<code>test_sliced_model_no_entity_leak</code>	<code>len(sliced.entities) == 0</code>	value_equality
<code>test_slicer.py</code>	<code>test_active_filter</code>	<code>cap.status == Status.ACTIVE</code>	value_equality
<code>test_slicer.py</code>	<code>test_slice_by_existing_layer</code>	<code>sliced.isinstance(ArchitectureModel)</code>	type_check
<code>test_slicer.py</code>	<code>test_slice_by_nonexistent_layer</code>	<code>len(sliced.entities.components) == 0</code>	value_equality

COMP-CORE-TYPES: `core.types`

- **F-Block:** F3
- **Description:** Type system (ArchitectureModel, Component, Capability, FunctionSignature, TestContract, etc.)
- **Files:** `src/architecture_model/core/types.py`

Signatures:

- `parse(value: str) --> RelationType | str @classmethod`
- `all_entity_ids() --> set[str] @property`
- `entity_count() --> int @property`
- `relationship_count() --> int @property`
- `to_dict() --> dict[str, Any]`

- to_yaml() --> str

Test Contracts (30):

Test File	Method	Assertion	Type
test_types.py	test_returns_dict	result isinstance dict	type_check
test_types.py	test_meta_fields	meta['schema_version'] == '1.0'	value_equality
test_types.py	test_status_is_string	actor['status'] == 'ACTIVE'	value_equality
test_types.py	test_actor_type_is_string	actor['type'] == 'human'	value_equality
test_types.py	test_interface_type_is_string	iface['type'] == 'REST'	value_equality
test_types.py	test_constraint_type_is_string	con['type'] == 'performance'	value_equality
test_types.py	test_priority_is_string	cap['priority'] == 'high'	value_equality
test_types.py	test_relationship_type_is_string	rel['type'] == 'realizes'	value_equality
test_types.py	test_strength_is_string	rel['strength'] == 'strong'	value_equality
test_types.py	test_from_to_values_compact	rel['from'] == 'comp-1'	value_equality
test_types.py	test_actor_fields	actor['id'] == 'actor-1'	value_equality
test_types.py	test_capability_fields	cap['f_block'] == 'FB-CodeGen'	value_equality
test_types.py	test_behavior_fields	beh['trigger'] == 'New project scan'	value_equality
test_types.py	test_interface_fields	iface['type'] == 'REST'	value_equality
test_types.py	test_constraint_fields	con['type'] == 'performance'	value_equality
test_types.py	test_layer_fields	layer['order'] == 2	value_equality
test_types.py	test_component_fields	comp['layer'] == 'layer-1'	value_equality
test_types.py	test_returns_string	result isinstance str	type_check
test_types.py	test_valid_yaml	parsed isinstance dict	type_check
test_types.py	test_yaml_matches_to_parsed	parsed == model.to_dict()	value_equality
test_types.py	test_round_trip_preserve_identity	original.entity_ids == rebuilt.all_entity_ids	value_equality
test_types.py	test_round_trip_preserve_identity_count	original.entity_count == rebuilt.entity_count	value_equality
test_types.py	test_round_trip_preserve_relationships	original.relationship_count == rebuilt.relationship_count	value_equality
test_types.py	test_round_trip_preserve_metadata	rebuilt.meta.schema_version == original.meta.schema_version	value_equality
test_types.py	test_round_trip_preserve_actor_details	rebuilt.actor_details == orig_actor.id	value_equality
test_types.py	test_round_trip_preserve_capability_details	rebuilt.cap_details == orig_rel.type	value_equality
test_types.py	test_round_trip_preserve_behavior_details	rebuilt.behavior_details == orig_trigger	value_equality
test_types.py	test_round_trip_preserve_component_details	rebuilt.component_details == orig_layer	value_equality
test_types.py	test_round_trip_minimal_rebuild	rebuilt.entity_count == original.entity_count	value_equality
test_types.py	test_yaml_round_trip	original.all_entity_ids == rebuilt.all_entity_ids	value_equality

COMP-CORE-VALIDATOR: core.validator

- **F-Block:** F3
- **Description:** Structural validation engine producing score 0-100
- **Files:** src/architecture_model/core/validator.py

Constants:

- ERROR = ERROR (Severity enum member)

- INFO = INFO (Severity enum member)

Signatures:

- error_count() --> int @property
- warning_count() --> int @property
- info_count() --> int @property
- is_valid() --> bool @property
- score() --> int @property
- summary() --> str
- validate_model(model: ArchitectureModel, strict: bool) --> ValidationResult

Test Contracts (11):

Test File	Method	Assertion	Type
test_validator.py	test_perfect_model_score	result.score == 100	value_equality
test_validator.py	test_score_deducts_10_percent	result.score == 90	value_equality
test_validator.py	test_score_deducts_2_percent_warning	result.score == 98	value_equality
test_validator.py	test_score_floors_at_zero	result.score == 0	value_equality
test_validator.py	test_info_does_not_affect_score	result.score == 100	value_equality
test_validator.py	test_duplicate_id_across_types	len(errors) == 0 Severity.ERROR	value_equality
test_validator.py	test_unique_ids_no_duplicates	len(errors) == 0	value_equality
test_validator.py	test_known_external_node_missing	len(errors) == 0	value_equality
test_validator.py	test_orphan_component_not_used	len(errors) == 0 Severity.INFO	value_equality
test_validator.py	test_orphan_behavior_preserved	len(errors) == 0 Severity.INFO	value_equality
test_validator.py	test_dormant_component_not_orphan	len(errors) == 0	value_equality

COMP-DECOMPOSE: decompose

- **F-Block:** F6
- **Description:** Model decomposition into per-F-block sub-models
- **Files:** src/architecture_model/orchestration/decompose.py

Signatures:

- decompose_model(project_root)
- write_sub_models(sub_models, output_dir)

Test Contracts (3):

Test File	Method	Assertion	Type
test_decompose.py	test_decompose_meta_links_to_parent_model	link.parent_model == '../..architecture-model.yaml'	value_equality
test_decompose.py	test_write_sub_models	data['meta']['parent_model'] == '../..architecture-mode...	value_equality
test_decompose.py	test_decompose_cli	ret == 0	value_equality

COMP-ENRICH: enrich

- **F-Block:** F6
- **Description:** Auto-enrichment of architecture models with AST-derived signatures, constants, and test contracts

- **Files:** src/architecture_model/orchestration/enrich.py

Signatures:

- `enrich_model(model: ArchitectureModel, project_root: Path) --> ArchitectureModel`

Test Contracts (2):

Test File	Method	Assertion	Type
test_enrich.py	test_enrich_skips_planned_signatures	<code>len(signatures) == 0</code>	value_equality
test_enrich.py	test_enrich_handles_missing_files	<code>len(signatures) == 0</code>	value_equality

COMP-EXTRACT: extract

- **F-Block:** F4
- **Description:** Extract architecture model from source code analysis
- **Files:** src/architecture_model/extract/from_code.py

Signatures:

- `extract_from_code(project_root: str | Path, config: ProjectConfig | None, manifest: dict | None) --> ArchitectureModel`

COMP-MANIFEST: manifest

- **F-Block:** F5
- **Description:** AST scanner, block discovery, metrics, interface detection, body hints, test analyzer

COMP-MANIFEST-BLOCKS: manifest.blocks

- **F-Block:** F5
- **Description:** Functional block discovery from source tree
- **Files:** src/architecture_model/manifest/blocks.py

Constants:

- `ACTIVE = ACTIVE` (Status/FileStatus enum member)

Signatures:

- `process_block(root: Path, block_id: str, block_def: dict, sub_block_configs: list | None) --> BlockManifest`

Test Contracts (21):

Test File	Method	Assertion	Type
test_blocks_typed.py	test_process_block_returns_block_manifest	<code>sub_block_manifest is instance of BlockManifest</code>	type_check
test_blocks_typed.py	test_process_block_backwards_compatible	<code>kwargs['concerns'] is instance of list</code>	type_check
test_blocks_typed.py	test_deprecated_process_block_is_instance_dict	<code>sub_block_manifest is instance of dict</code>	type_check
test_blocks_typed.py	test_sub_function_entry_filenames_and_strings	<code>isinstance(strings, list)</code>	type_check
test_manifest_types.py	test_module_info_creation	<code>file == 'src/foo.py'</code>	value_equality
test_manifest_types.py	test_module_status_value	<code>ModuleStatus.ACTIVE.value == 'active'</code>	value_equality
test_manifest_types.py	test_scan_report_success_rate	<code>success_rate == 0.9</code>	value_equality

Test File	Method	Assertion	Type
test_manifest_types.py	test_scan_report_empty	r.success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	source == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict_backwards_compat	dict == instance dict	type_check
test_manifest_types.py	test_module_info_to_dict	['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_point	dict == 'F1.1'	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] is instance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] is instance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] is instance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	block_def['subfunctions'] is instance list	type_check
test_manifest.py	test_has_modules	generated['modules'] is instance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] is instance list	type_check
test_manifest.py	test_returns_non_empty_results	result is instance str	type_check

COMP-MANIFEST-BODY-HINTS: manifest.body_hints

- **F-Block:** F5
- **Description:** Function body hint extraction for trivial implementations
- **Files:** src/architecture_model/manifest/body_hints.py

Constants:

- TRIVIAL = trivial (BodyComplexity enum member)
- SHORT = short (BodyComplexity enum member)
- COMPLEX = complex (BodyComplexity enum member)
- ACTIVE = ACTIVE (Status/FileStatus enum member)

Signatures:

- classify_function(source: str, func_name: str) --> BodyComplexity
- extract_body_hint(source: str, func_name: str, class_name: str | None) --> str
- extract_file_hints(filepath: Path, include_private: bool) --> list[FunctionSignature]

Test Contracts (29):

Test File	Method	Assertion	Type
test_body_hints.py	test_trivial_single_statement	classify_function(source, 'greet') == BodyComplexity.TRIVIAL	value_equality
test_body_hints.py	test_trivial_with_docstring	classify_function(source, 'greet') == BodyComplexity.TRIVIAL	value_equality
test_body_hints.py	test_short_two_statements	classify_function(source, 'add') == BodyComplexity.SHORT	value_equality
test_body_hints.py	test_short_five_statements	classify_function(source, 'process') == BodyComplexity.SHORT	value_equality

Test File	Method	Assertion	Type
test_body_hints.py	test_complex_six_statements	classify_function(source, 'complex_func') == BodyComplexity.C...	value_equality
test_body_hints.py	test_complex_with_docs	classify_function(source, 'big_func') == BodyComplexity.C...	value_equality
test_body_hints.py	test_function_not_found	raises ValueError	raises
test_body_hints.py	test_trivial_returns_exact_body	'return 42'	value_equality
test_body_hints.py	test_class_method_extraction	'return a + b'	value_equality
test_body_hints.py	test_body_hint_populated	sig.body_hint == 'return 42'	value_equality
test_body_hints.py	test_params_populated	sig.returns == 'float'	value_equality
test_body_hints.py	test_returns_populated	sig.returns == 'str'	value_equality
test_manifest_types.py	test_module_info_creation	on.file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty	success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	source == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict_body	m.metrics['body'] is instance dict	type_check
test_manifest_types.py	test_module_info_to_dict	['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_dict	['id'] == 'F1.1'	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] is instance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] is instance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] is instance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	generated['subfunctions'] is instance list	type_check
test_manifest.py	test_has_modules	generated['modules'] is instance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] is instance list	type_check
test_manifest.py	test_returns_non_empty_result	result is instance str	type_check

COMP-MANIFEST-GENERATOR: manifest.generator

- **F-Block:** F5
- **Description:** Top-level manifest generation orchestrator
- **Files:** src/architecture_model/manifest/generator.py, src/architecture_model/manifest/display.py, src/architecture_model/manifest/slicers.py

Constants:

- ACTIVE = ACTIVE (Status/FileStatus enum member)

Signatures:

- generate_manifest(project_root: Path, config: Optional[Any]) --> Manifest
- load_or_generate_manifest(project_root: Path, output_dir: Path | None) --> dict[str, Any]
- print_summary(manifest: dict[str, Any]) --> None

- `get_manifest_slice(manifest: Manifest | dict[str, Any], artifact_name: str) --> str`

Test Contracts (21):

Test File	Method	Assertion	Type
test_generator_typed.py	test_generate_manifest	result.isinstance(Manifest)	type_check
test_generator_typed.py	test_generate_manifest	report.isinstance(Manifest)	type_check
test_generator_typed.py	test_scan_report_tracks	report.files_succeeded == report.files_attempted	value_equality
test_generator_typed.py	test_load_or_generate_manifest	result.isinstance(dict)	type_check
test_manifest_types.py	test_module_info_creation	m.file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty_r	success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	source == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict	dict[metrics] is instance dict	type_check
test_manifest_types.py	test_module_info_to_dict	['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_dict	['id'] == 'F1.1'	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] is instance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] is instance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] is instance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	['subfunctions'] is instance list	type_check
test_manifest.py	test_has_modules	generated['modules'] is instance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] is instance list	type_check
test_manifest.py	test_returns_non_empty_manifest	result is instance str	type_check

COMP-MANIFEST-INTERFACES: manifest.interfaces

- **F-Block:** F5
- **Description:** Interface detection from public API surface
- **Files:** src/architecture_model/manifest/interfaces.py

Constants:

- ACTIVE = ACTIVE (Status/FileStatus enum member)

Signatures:

- `derive_interfaces(modules: list[ModuleInfo], root: Path) --> list[InterfaceEdge]`

Test Contracts (20):

Test File	Method	Assertion	Type
test_interfaces_typed.py	test_derive_interfaces	result[0].source == 'pkg/a.py'	value_equality
test_interfaces_typed.py	test_derive_interfaces	len(set_difference)	value_equality
test_interfaces_typed.py	test_derive_interfaces	len(uniques) == 1	value_equality

Test File	Method	Assertion	Type
test_manifest_types.py	test_module_info_creation	m.file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	r.success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty_r	r.success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	r.source == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict_backwards	m.metrics is instance dict	type_check
test_manifest_types.py	test_module_info_to_dict	d['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	d['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_dict	d['F1.1'] == 'F1.1'	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] is instance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] is instance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] is instance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	block_def['subfunctions'] is instance list	type_check
test_manifest.py	test_has_modules	generated['modules'] is instance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] is instance list	type_check
test_manifest.py	test_returns_non_empty_results	results is instance str	type_check

COMP-MANIFEST-METRICS: manifest.metrics

- **F-Block:** F5
- **Description:** Code metrics computation (LOC, complexity, etc.)
- **Files:** src/architecture_model/manifest/metrics.py

Constants:

- ACTIVE = ACTIVE (Status/FileStatus enum member)

Signatures:

- compute_metrics(root: Path, config: Optional[Any]) --> MetricsResult

Test Contracts (19):

Test File	Method	Assertion	Type
test_metrics_typed.py	test_compute_metrics_results	results is instance MetricsResult	type_check
test_metrics_typed.py	test_compute_metrics_backward_compat	backward_compat	type_check
test_manifest_types.py	test_module_info_creation	m.file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	r.success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty_r	r.success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	r.source == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict_backwards	m.metrics is instance dict	type_check

Test File	Method	Assertion	Type
test_manifest_types.py	test_module_info_to_dict	dict['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	dict['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_to_dict	dict['F1.1'] == 'F1.1'	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] isinstance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] isinstance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] isinstance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	block_def['subfunctions'] isinstance list	type_check
test_manifest.py	test_has_modules	generated['modules'] isinstance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] isinstance list	type_check
test_manifest.py	test_returns_non_empty_result	result isinstance str	type_check

COMP-MANIFEST-SCANNER: manifest.scanner

- **F-Block:** F5
- **Description:** AST-based source code scanner
- **Files:** src/architecture_model/manifest/scanner.py

Constants:

- ACTIVE = ACTIVE (Status/FileStatus enum member)
- MISSING = missing (FileStatus enum member)

Signatures:

- scan_file(root: Path, filepath: Path) --> ModuleInfo

Test Contracts (21):

Test File	Method	Assertion	Type
test_scanner_typed.py	test_scan_file_returns_module_info	result isinstance ModuleInfo	type_check
test_scanner_typed.py	test_scan_file_parse_error	result.status == ModuleStatus.MISSING	value_equality
test_scanner_typed.py	test_deprecated_scan_files_returns_dict	result isinstance dict	type_check
test_scanner_typed.py	test_deprecated_collect_files	len(files) == 2	value_equality
test_manifest_types.py	test_module_info_creation	file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty_rate	success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	resource == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict	dict['metrics'] isinstance dict	type_check
test_manifest_types.py	test_module_info_to_dict	dict['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	dict['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_to_dict	dict['F1.1'] == 'F1.1'	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] isinstance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] isinstance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] isinstance dict	type_check

Test File	Method	Assertion	Type
test_manifest.py	test_functional_blocks_have_subfunctions	have_subfunctions isinstance list	type_check
test_manifest.py	test_has_modules	generated['modules'] isinstance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] isinstance list	type_check
test_manifest.py	test_returns_non_empty_result	isinstance str	type_check

COMP-MANIFEST-TEST-ANALYZER: manifest.test_analyzer

- **F-Block:** F5
- **Description:** Test contract extraction from test files
- **Files:** src/architecture_model/manifest/test_analyzer.py

Constants:

- `_EXCLUDED_MODULES = frozenset({'unittest', 'pytest', 'sys', 'os', 'io', 're', 'math', 'collections', 'itertools', 'functools', 'pathlib', 'typing', 'tempfile', 'shutil', 'json', 'copy', 'contextlib', 'textwrap', 'unittest.mock', 'mock'})`
- `_ESCAPE_CODE_RE = re.compile('\\\\\\\\(?:033|x1b)\\\\\\\\[(\\\\\\\\d+)m')`
- `ACTIVE = ACTIVE` (Status/FileStatus enum member)

Signatures:

- `analyze_test_file(test_file: Path) --> TestAnalysisResult`
- `extract_constants_from_contracts(contracts: list[TestContract]) --> list[Constant]`

Test Contracts (24):

Test File	Method	Assertion	Type
test_test_analyzer.py	test_result_has_expected_fields	TestAnalysisResult	type_check
test_test_analyzer.py	test_assertEqual_produces_instance_of_TestContract	TestContract	type_check
test_test_analyzer.py	test_test_count	result.test_count == 4	value_equality
test_test_analyzer.py	test_constant_values_are_black_value_codes	black_value_codes	value_equality
test_test_analyzer.py	test_extract_constants_from_constants_standalone	from_constants_standalone	value_equality
test_test_analyzer.py	test_fore_black_value	black.value == '30'	value_equality
test_test_analyzer.py	test_fore_reset_value	resets[0].value == '39'	value_equality
test_manifest_types.py	test_module_info_creation	file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty_rate	success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	resource == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict_backwards	dict instance dict	type_check
test_manifest_types.py	test_module_info_to_dict	['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_dict	['F1.1']	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] isinstance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] isinstance list	type_check

Test File	Method	Assertion	Type
test_manifest.py	test_has_metrics	generated['metrics'] isinstance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	generated['subfunctions'] isinstance list	type_check
test_manifest.py	test_has_modules	generated['modules'] isinstance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] isinstance list	type_check
test_manifest.py	test_returns_non_empty_result	result is instance str	type_check

COMP-MANIFEST-TYPES: manifest.types

- **F-Block:** F5
- **Description:** Manifest data types (Manifest, ModuleInfo, FunctionalBlock)
- **Files:** src/architecture_model/manifest/types.py

Constants:

- ACTIVE = ACTIVE (Status/FileStatus enum member)

Signatures:

- to_dict() --> dict[str, Any]
- success_rate() --> float @property
- log_summary() --> None

Test Contracts (17):

Test File	Method	Assertion	Type
test_manifest_types.py	test_module_info_creation	on file == 'src/foo.py'	value_equality
test_manifest_types.py	test_module_status_value	ModuleStatus.ACTIVE.value == 'active'	value_equality
test_manifest_types.py	test_scan_report_success_rate	success_rate == 0.9	value_equality
test_manifest_types.py	test_scan_report_empty_r	success_rate == 1.0	value_equality
test_manifest_types.py	test_interface_edge_creation	source == 'a/b.py'	value_equality
test_manifest_types.py	test_manifest_creation	m.metrics.values['total_python_files'] == 10	value_equality
test_manifest_types.py	test_manifest_to_dict_block_def	block_def is instance dict	type_check
test_manifest_types.py	test_module_info_to_dict	['status'] == 'active'	value_equality
test_manifest_types.py	test_block_manifest_to_dict	['name'] == 'Core'	value_equality
test_manifest_types.py	test_sub_function_entry_dict	['F1.1']	value_equality
test_manifest.py	test_each_block_has_dirs	block_def['dirs'] is instance list	type_check
test_manifest.py	test_each_block_has_files	block_def['files'] is instance list	type_check
test_manifest.py	test_has_metrics	generated['metrics'] isinstance dict	type_check
test_manifest.py	test_functional_blocks_have_subfunctions	generated['subfunctions'] isinstance list	type_check
test_manifest.py	test_has_modules	generated['modules'] isinstance list	type_check
test_manifest.py	test_has_interfaces	generated['interfaces'] isinstance list	type_check
test_manifest.py	test_returns_non_empty_result	result is instance str	type_check

COMP-PROFILES: profiles

- **F-Block:** F7
- **Description:** Domain profile system (software, controls, mechanical, electrical)
- **Files:** src/architecture_model/profiles/schema.py, src/architecture_model/profiles/init.py, src/architecture_model/profiles/

Constants:

- PROFILES_DIR = Path(__file__).parent / 'builtins'

Signatures:

- from_dict(data: dict[str, Any]) --> DomainProfile @classmethod
- get_extended_values(enum_name: str) --> list[str]
- load_profile(name_or_path: str) --> DomainProfile

COMP-SPEC: spec

- **F-Block:** F8
- **Description:** JSON Schema definitions for architecture model validation
- **Files:** src/architecture_model/spec/init.py

COMP-UTILS: utils

- **F-Block:** F9
- **Description:** File discovery, exclusion patterns, and shared helper functions
- **Files:** src/architecture_model/utils/discovery.py, src/architecture_model/utils/init.py

Signatures:

- is_excluded_dir(path: Path) --> bool
- collect_py_files(directory: Path, recursive: bool, exclude_init: bool) --> list[Path]
- discover_source_files(project_root: Path) --> list[Path]
- discover_test_files(project_root: Path) --> list[Path]

Test Contracts (7):

Test File	Method	Assertion	Type
test_discovery.py	test_collect_py_files_excludes_cache	len(result) == 2	value_equality
test_discovery.py	test_collect_py_files_recursive	len(result) == 2	value_equality
test_discovery.py	test_collect_py_files_nonrecursive	len(result) == 1	value_equality
test_discovery.py	test_collect_py_files_excludes_init	len(result) == 1	value_equality
test_discovery.py	test_collect_py_files_nonexistent_dir	len(result) == 0	value_equality
test_discovery.py	test_discover_test_files	len(result) == 2	value_equality
test_discovery.py	test_discover_test_files_by_rule	len(result) == 2	value_equality

Relationships

Type	From	To
contains	L-APP	COMP-CORE
contains	L-APP	COMP-MANIFEST
contains	L-APP	COMP-CLI
contains	L-APP	COMP-CONFIG
contains	L-APP	COMP-SPEC

Type	From	To
contains	L-APP	COMP-EXTRACT
contains	L-APP	COMP-PROFILES
contains	L-APP	COMP-UTILS
contains	L-APP	COMP-ENRICH
contains	COMP-CORE	COMP-CORE-PARSER
contains	COMP-CORE	COMP-CORE-VALIDATOR
contains	COMP-CORE	COMP-CORE-SLICER
contains	COMP-CORE	COMP-CORE-DIFFER
contains	COMP-CORE	COMP-CORE-MERGER
contains	COMP-CORE	COMP-CORE-DECOMPOSER
contains	COMP-CORE	COMP-CORE-TYPES
contains	COMP-MANIFEST	COMP-MANIFEST-SCANNER
contains	COMP-MANIFEST	COMP-MANIFEST-BLOCKS
contains	COMP-MANIFEST	COMP-MANIFEST-METRICS
contains	COMP-MANIFEST	COMP-MANIFEST-INTERFACES
contains	COMP-MANIFEST	COMP-MANIFEST-BODY-HINTS
contains	COMP-MANIFEST	COMP-MANIFEST-TEST-ANALYZER
contains	COMP-MANIFEST	COMP-MANIFEST-GENERATOR
contains	COMP-MANIFEST	COMP-MANIFEST-TYPES
realizes	COMP-CORE	CAP-F1
realizes	COMP-CORE	CAP-F3
realizes	COMP-MANIFEST	CAP-F2
realizes	COMP-CLI	CAP-F4
realizes	COMP-CONFIG	CAP-F5
realizes	COMP-SPEC	CAP-F6
realizes	COMP-EXTRACT	CAP-F7
realizes	COMP-PROFILES	CAP-F8
realizes	COMP-UTILS	CAP-F9
realizes	COMP-ENRICH	CAP-F10
exposes	COMP-CLI	IF-CLI
exposes	COMP-CORE-PARSER	IF-PARSE-API
exposes	COMP-CORE-VALIDATOR	IF-VALIDATE-API
exposes	COMP-CORE-SLICER	IF-SLICER-API
exposes	COMP-MANIFEST-GENERATOR	IF-MANIFEST-API
exposes	COMP-PROFILES	IF-PROFILE-API
exposes	COMP-ENRICH	IF-ENRICH-API
constrained-by	COMP-CORE	CON-SCHEMA
constrained-by	COMP-SPEC	CON-SCHEMA
constrained-by	COMP-CORE	CON-NO-ORPHANS
depends-on	COMP-CLI	COMP-CORE
depends-on	COMP-CLI	COMP-CONFIG
depends-on	COMP-CLI	COMP-MANIFEST
depends-on	COMP-CLI	COMP-ENRICH
depends-on	COMP-CORE	COMP-CONFIG
depends-on	COMP-CORE	COMP-SPEC
depends-on	COMP-CORE	COMP-PROFILES
depends-on	COMP-MANIFEST	COMP-CONFIG
depends-on	COMP-MANIFEST	COMP-UTILS
depends-on	COMP-EXTRACT	COMP-CORE
depends-on	COMP-ENRICH	COMP-CORE
depends-on	COMP-ENRICH	COMP-MANIFEST
depends-on	COMP-CORE-PARSER	COMP-CORE-TYPES

Type	From	To
depends-on	COMP-CORE-VALIDATOR	COMP-CORE-TYPES
depends-on	COMP-CORE-VALIDATOR	COMP-PROFILES
depends-on	COMP-CORE-SLICER	COMP-CORE-TYPES
depends-on	COMP-CORE-DIFFER	COMP-CORE-TYPES
depends-on	COMP-CORE-MERGER	COMP-CORE-TYPES
depends-on	COMP-CORE-MERGER	COMP-UTILS
depends-on	COMP-CORE-DECOMPOSER	COMP-CORE-TYPES
depends-on	COMP-CORE-DECOMPOSER	COMP-UTILS
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-SCANNER
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-BLOCKS
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-METRICS
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-INTERFACES
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-BLOCKS	COMP-MANIFEST-SCANNER
depends-on	COMP-MANIFEST-BLOCKS	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-BLOCKS	COMP-UTILS
depends-on	COMP-MANIFEST-SCANNER	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-METRICS	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-INTERFACES	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-BODY-HINTS	COMP-CORE-TYPES
depends-on	COMP-MANIFEST-TEST-ANALYZER	COMP-CORE-TYPES
traces-to	COMP-CLI	BEH-INIT
traces-to	COMP-CLI	BEH-VALIDATE
traces-to	COMP-CLI	BEH-MANIFEST
traces-to	COMP-CLI	BEH-ENRICH
traces-to	COMP-CORE-PARSER	BEH-VALIDATE
traces-to	COMP-CORE-VALIDATOR	BEH-VALIDATE
traces-to	COMP-MANIFEST-GENERATOR	BEH-MANIFEST
traces-to	COMP-ENRICH	BEH-ENRICH
traces-to	COMP-EXTRACT	BEH-EXTRACT
consumes	ACT-DEV	IF-CLI
consumes	ACT-LLM	IF-PARSE-API
consumes	ACT-LLM	IF-SLICER-API
consumes	ACT-LLM	IF-MANIFEST-API
depends-on	COMP-DECOMPOSE	COMP-CORE
depends-on	COMP-DECOMPOSE	COMP-CONFIG

Sub-Model F1: Cli

- **Validation Score:** 90/100
- **Refines:** COMP-CLI
- **Components:** 1
- **Capabilities:** 1
- **Interfaces:** 1
- **Behaviors:** 4
- **Constraints:** 0
- **Relationships:** 10

Capabilities:

- CAP-F4: CLI Operations (F-Block F1)

Interfaces:

- IF-CLI: CLI Interface

Behaviors:

- BEH-INIT: Project Initialization
- BEH-VALIDATE: Model Validation
- BEH-MANIFEST: Manifest Generation
- BEH-ENRICH: Auto-Enrichment

Components:

ID	Name	Sigs	Const	Contracts
COMP-CLI	cli	1	0	0

Relationships:

Type	From	To
realizes	COMP-CLI	CAP-F4
exposes	COMP-CLI	IF-CLI
depends-on	COMP-CLI	COMP-CORE
depends-on	COMP-CLI	COMP-CONFIG
depends-on	COMP-CLI	COMP-MANIFEST
depends-on	COMP-CLI	COMP-ENRICH
traces-to	COMP-CLI	BEH-INIT
traces-to	COMP-CLI	BEH-VALIDATE
traces-to	COMP-CLI	BEH-MANIFEST
traces-to	COMP-CLI	BEH-ENRICH

Sub-Model F2: Config

- **Validation Score:** 90/100
- **Refines:** COMP-CONFIG
- **Components:** 1
- **Capabilities:** 1
- **Interfaces:** 0
- **Behaviors:** 0
- **Constraints:** 0
- **Relationships:** 5

Capabilities:

- CAP-F5: Configuration Management (F-Block F2)

Components:

ID	Name	Sigs	Const	Contracts
COMP-CONFIG	config	15	1	1

Relationships:

Type	From	To
realizes	COMP-CONFIG	CAP-F5
depends-on	COMP-CLI	COMP-CONFIG
depends-on	COMP-CORE	COMP-CONFIG
depends-on	COMP-MANIFEST	COMP-CONFIG
depends-on	COMP-DECOMPOSE	COMP-CONFIG

Sub-Model F3: Core

- **Validation Score:** 74/100
- **Refines:** COMP-CORE
- **Components:** 8
- **Capabilities:** 2
- **Interfaces:** 3
- **Behaviors:** 1
- **Constraints:** 2
- **Relationships:** 34

Capabilities:

- CAP-F1: Model Parsing & Validation (F-Block F3)
- CAP-F3: Model Slicing & Diffing (F-Block F3)

Interfaces:

- IF-PARSE-API: Parser API
- IF-VALIDATE-API: Validator API
- IF-SLICER-API: Slicer API

Behaviors:

- BEH-VALIDATE: Model Validation

Constraints:

- CON-SCHEMA: Schema Compliance
- CON-NO-ORPHANS: No Orphaned Entities

Components:

ID	Name	Sigs	Const	Contracts
COMP-CORE	core	0	0	0
COMP-CORE-PARSER	core.parser	4	1	11
COMP-CORE-VALIDATOR	core.validator	7	2	11
COMP-CORE-SLICER	core.slicer	4	1	12
COMP-CORE-DIFFER	core.differ	8	0	0
COMP-CORE-MERGER	core.merger	8	5	2
COMP-CORE-DECOMPOSER	core.decomposer	5	1	15
COMP-CORE-TYPES	core.types	6	0	30

Relationships:

Type	From	To
contains	COMP-CORE	COMP-CORE-PARSER
contains	COMP-CORE	COMP-CORE-VALIDATOR
contains	COMP-CORE	COMP-CORE-SLICER
contains	COMP-CORE	COMP-CORE-DIFFER
contains	COMP-CORE	COMP-CORE-MERGER
contains	COMP-CORE	COMP-CORE-DECOMPOSER
contains	COMP-CORE	COMP-CORE-TYPES
realizes	COMP-CORE	CAP-F1
realizes	COMP-CORE	CAP-F3
exposes	COMP-CORE-PARSER	IF-PARSE-API

Type	From	To
exposes	COMP-CORE-VALIDATOR	IF-VALIDATE-API
exposes	COMP-CORE-SLICER	IF-SLICER-API
constrained-by	COMP-CORE	CON-SCHEMA
constrained-by	COMP-CORE	CON-NO-ORPHANS
depends-on	COMP-CLI	COMP-CORE
depends-on	COMP-CORE	COMP-CONFIG
depends-on	COMP-CORE	COMP-SPEC
depends-on	COMP-CORE	COMP-PROFILES
depends-on	COMP-EXTRACT	COMP-CORE
depends-on	COMP-ENRICH	COMP-CORE
depends-on	COMP-CORE-PARSER	COMP-CORE-TYPES
depends-on	COMP-CORE-VALIDATOR	COMP-CORE-TYPES
depends-on	COMP-CORE-VALIDATOR	COMP-PROFILES
depends-on	COMP-CORE-SLICER	COMP-CORE-TYPES
depends-on	COMP-CORE-DIFFER	COMP-CORE-TYPES
depends-on	COMP-CORE-MERGER	COMP-CORE-TYPES
depends-on	COMP-CORE-MERGER	COMP-UTILS
depends-on	COMP-CORE-DECOMPOSER	COMP-CORE-TYPES
depends-on	COMP-CORE-DECOMPOSER	COMP-UTILS
depends-on	COMP-MANIFEST-BODY-HINTS	COMP-CORE-TYPES
depends-on	COMP-MANIFEST-TEST-ANALYZER	COMP-CORE-TYPES
traces-to	COMP-CORE-PARSER	BEH-VALIDATE
traces-to	COMP-CORE-VALIDATOR	BEH-VALIDATE
depends-on	COMP-DECOMPOSE	COMP-CORE

Sub-Model F4: Extract

- **Validation Score:** 96/100
- **Refines:** COMP-EXTRACT
- **Components:** 1
- **Capabilities:** 1
- **Interfaces:** 0
- **Behaviors:** 1
- **Constraints:** 0
- **Relationships:** 3

Capabilities:

- CAP-F7: Model Extraction (F-Block F4)

Behaviors:

- BEH-EXTRACT: Model Extraction from Code

Components:

ID	Name	Sigs	Const	Contracts
COMP-EXTRACT	extract	1	0	0

Relationships:

Type	From	To
realizes	COMP-EXTRACT	CAP-F7
depends-on	COMP-EXTRACT	COMP-CORE
traces-to	COMP-EXTRACT	BEH-EXTRACT

Sub-Model F5: Manifest

- **Validation Score:** 84/100
- **Refines:** COMP-MANIFEST
- **Components:** 9
- **Capabilities:** 1
- **Interfaces:** 1
- **Behaviors:** 1
- **Constraints:** 0
- **Relationships:** 28

Capabilities:

- CAP-F2: Reality Manifest Generation (F-Block F5)

Interfaces:

- IF-MANIFEST-API: Manifest API

Behaviors:

- BEH-MANIFEST: Manifest Generation

Components:

ID	Name	Sigs	Const	Contracts
COMP-MANIFEST	manifest	0	0	0
COMP-MANIFEST-SCANNER	manifest.scanner	1	2	21
COMP-MANIFEST-BLOCKS	manifest.blocks	1	1	21
COMP-MANIFEST-METRICS	manifest.metrics	1	1	19
COMP-MANIFEST-INTERFACES	manifest.interfaces	1	1	20
COMP-MANIFEST-BODY-HINTS	manifest.body_hints	3	4	29
COMP-MANIFEST-TEST-ANALYZER	manifest.test_analyzer	2	3	24
COMP-MANIFEST-GENERATOR	manifest.generator	4	1	21
COMP-MANIFEST-TYPES	manifest.types	3	1	17

Relationships:

Type	From	To
contains	COMP-MANIFEST	COMP-MANIFEST-SCANNER
contains	COMP-MANIFEST	COMP-MANIFEST-BLOCKS
contains	COMP-MANIFEST	COMP-MANIFEST-METRICS
contains	COMP-MANIFEST	COMP-MANIFEST-INTERFACES
contains	COMP-MANIFEST	COMP-MANIFEST-BODY-HINTS
contains	COMP-MANIFEST	COMP-MANIFEST-TEST-ANALYZER
contains	COMP-MANIFEST	COMP-MANIFEST-GENERATOR
contains	COMP-MANIFEST	COMP-MANIFEST-TYPES
realizes	COMP-MANIFEST	CAP-F2
exposes	COMP-MANIFEST-GENERATOR	IF-MANIFEST-API
depends-on	COMP-CLI	COMP-MANIFEST
depends-on	COMP-MANIFEST	COMP-CONFIG
depends-on	COMP-MANIFEST	COMP-UTILS
depends-on	COMP-ENRICH	COMP-MANIFEST
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-SCANNER
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-BLOCKS

Type	From	To
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-METRICS
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-INTERFACES
depends-on	COMP-MANIFEST-GENERATOR	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-BLOCKS	COMP-MANIFEST-SCANNER
depends-on	COMP-MANIFEST-BLOCKS	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-BLOCKS	COMP-UTILS
depends-on	COMP-MANIFEST-SCANNER	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-METRICS	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-INTERFACES	COMP-MANIFEST-TYPES
depends-on	COMP-MANIFEST-BODY-HINTS	COMP-CORE-TYPES
depends-on	COMP-MANIFEST-TEST-ANALYZER	COMP-CORE-TYPES
traces-to	COMP-MANIFEST-GENERATOR	BEH-MANIFEST

Sub-Model F6: Orchestration

- **Validation Score:** 88/100
- **Components:** 2
- **Capabilities:** 1
- **Interfaces:** 1
- **Behaviors:** 1
- **Constraints:** 0
- **Relationships:** 8

Capabilities:

- CAP-F10: Auto-Enrichment (F-Block F6)

Interfaces:

- IF-ENRICH-API: Enrichment API

Behaviors:

- BEH-ENRICH: Auto-Enrichment

Components:

ID	Name	Sigs	Const	Contracts
COMP-ENRICH	enrich	1	0	2
COMP-DECOMPOSE	decompose	2	0	3

Relationships:

Type	From	To
realizes	COMP-ENRICH	CAP-F10
exposes	COMP-ENRICH	IF-ENRICH-API
depends-on	COMP-CLI	COMP-ENRICH
depends-on	COMP-ENRICH	COMP-CORE
depends-on	COMP-ENRICH	COMP-MANIFEST
traces-to	COMP-ENRICH	BEH-ENRICH
depends-on	COMP-DECOMPOSE	COMP-CORE
depends-on	COMP-DECOMPOSE	COMP-CONFIG

Sub-Model F7: Profiles

- **Validation Score:** 94/100
- **Refines:** COMP-PROFILES
- **Components:** 1
- **Capabilities:** 1
- **Interfaces:** 1
- **Behaviors:** 0
- **Constraints:** 0
- **Relationships:** 4

Capabilities:

- CAP-F8: Domain Profiles (F-Block F7)

Interfaces:

- IF-PROFILE-API: Profile API

Components:

ID	Name	Sigs	Const	Contracts
COMP-PROFILES	profiles	3	1	0

Relationships:

Type	From	To
realizes	COMP-PROFILES	CAP-F8
exposes	COMP-PROFILES	IF-PROFILE-API
depends-on	COMP-CORE	COMP-PROFILES
depends-on	COMP-CORE-VALIDATOR	COMP-PROFILES

Sub-Model F8: Spec

- **Validation Score:** 96/100
- **Refines:** COMP-SPEC
- **Components:** 1
- **Capabilities:** 1
- **Interfaces:** 0
- **Behaviors:** 0
- **Constraints:** 1
- **Relationships:** 3

Capabilities:

- CAP-F6: Schema Specification (F-Block F8)

Constraints:

- CON-SCHEMA: Schema Compliance

Components:

ID	Name	Sigs	Const	Contracts
COMP-SPEC	spec	0	0	0

Relationships:

Type	From	To
realizes	COMP-SPEC	CAP-F6
constrained-by	COMP-SPEC	CON-SCHEMA
depends-on	COMP-CORE	COMP-SPEC

Sub-Model F9: Utils

- **Validation Score:** 90/100
- **Refines:** COMP-UTILS
- **Components:** 1
- **Capabilities:** 1
- **Interfaces:** 0
- **Behaviors:** 0
- **Constraints:** 0
- **Relationships:** 5

Capabilities:

- CAP-F9: Shared Utilities (F-Block F9)

Components:

ID	Name	Sigs	Const	Contracts
COMP-UTILS	utils	4	0	7

Relationships:

Type	From	To
realizes	COMP-UTILS	CAP-F9
depends-on	COMP-MANIFEST	COMP-UTILS
depends-on	COMP-CORE-MERGER	COMP-UTILS
depends-on	COMP-CORE-DECOMPOSER	COMP-UTILS
depends-on	COMP-MANIFEST-BLOCKS	COMP-UTILS